

September 11, 2024



FISSURE Capabilities

Core Functionality Overview

Christopher Poore
Senior Reverse Engineer
Assured Information Security, Inc.
poorec@ainfosec.com

Outline

- Reference Material: README, User Manual
- Operational Flow with Current Limitations
- Supported Hardware
- RF Protocols
- Attack Categories
- Attack Examples
- Trigger Examples



Reference Material

README, User Manual

- README: <https://github.com/ainfosec/FISSURE>
- User Manual: <https://fissure.readthedocs.io/en/latest/>
 - Installation
 - Requirements, Cloning, Installer, Remote Sensor Node Installation, Uninstalling, Local Dashboard Usage, Remote Sensor Node Usage, Known Conflicts, Third-Party Software, Third-Party Software Versions
 - Hardware
 - Supported Peripherals, Supported Sensor Node Hardware, Notes
 - Components
 - Overview, Dashboard, Target Signal Identification, Protocol Discovery, HIPRFISR/Central Hub, Sensor Nodes
 - Operation
 - Start-Up Procedures, Menu Items, Automation Tab, TSI Tab, PD Tab, Attack Tab, IQ Data Tab, Archive Tab, Sensor Nodes Tab, Library Tab, Log Tab, Status Bar, Triggers
 - Development
 - Built With, Modifying Dashboard, Style Sheets, Adding Custom Options, Attack Flow Graphs, Attack Python Scripts, Inspection Flow Graphs, Creating Triggers
 - About
 - Credits



FISSURE as a Disrupter Tool

Operational Flow

For Disrupter Scenarios

- Choose deployment scenario and target application
- Install FISSURE software on hardware platform
- Connect hardware for radios, sensors, amplifiers, antennas, etc.
- Decide on autonomous or interactive operation
- Assign triggers for autorun playlists using available sensors
- Add single-stage and multi-stage attacks/scripts items to autorun playlists for radio hardware
- Complete mission, download any captured data



Operational Flow

Current Limitations for Disrupter Scenarios

- In an experimental phase, not tested for SWaP
- Practical RF capabilities need to be developed
- Networking for the remote nodes is very limited
 - No communications from node to node
 - No live viewing of data
 - Data transfer is slow and needs enhancements
- Needs improved error handling and status messaging
- Requires extra amplification hardware for long-distance transmission

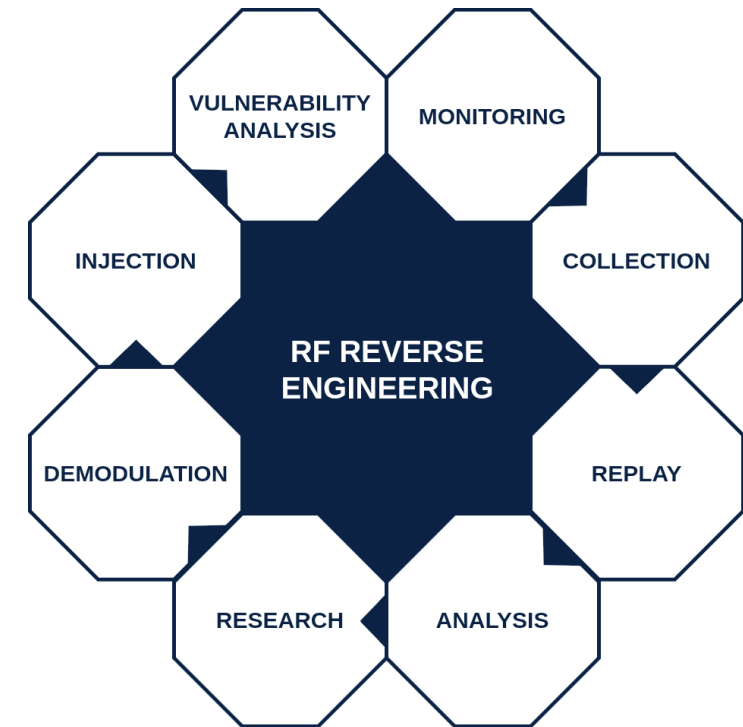


FISSURE as a RF Reverse Engineering Tool

Operational Flow

Overview

- Provide visualizations, tools, algorithms, AI/ML techniques for supporting each phase in the RF Reverse Engineering process
- Target Signal Identification Tab – Obtain signals of interest with some preliminary analysis
- Protocol Discovery Tab – Identify RF protocol parameters for demodulation and perform analysis of the bitstream, expand FISSURE library with protocol information
- Attack Tab – Execute single-stage, multi-stage, or fuzzing attacks using third-party tools, Python scripts, GNU Radio flow graphs, and crafted packets from the FISSURE library
- IQ Data Tab – Perform manual analysis of signal data through recording, playback, live inspection, data manipulation, filtering, and viewing
- Archive Tab – Download signal file collections from the AIS online archive, build signal playlists, construct datasets with randomizations for machine learning training
- Sensor Nodes Tab – Construct autorun playlists from single-stage and multi-stage attacks to run on remote sensor nodes, perform file transfer functions between the remote sensor nodes and the local workstation
- Library Tab – Browse/search the FISSURE library, view saved user images, add/remove: protocols, modulation types, packet types, signals of interest, statistics, demodulation flow graphs, attacks
- Log Tab – View and filter the system log, set logging levels, create session notes
- Menu Items – Browse options, customize color schemes, open standalone flow graphs, experiment with third-party tools, view reference material and lessons, access help items and local user manual



Operational Flow

Target Signal Identification

- Target Signal Identification Tab – Obtain signals of interest with some preliminary analysis
 - Detection – Scan for activity, identify frequencies of interest
 - Signal Conditioning – Isolate signals from large streams of IQ data (recordings or live data capture)
 - Feature Extraction – Extract statistical features from isolated signals for machine learning models
 - Signal Classification – Develop machine learning models, test against unknown signals, identify protocols/emitters, compare models, establish confidence
 - Store signals, features, classification results as SOIs in database



Operational Flow

Target Signal Identification – Current Limitations

- Target Signal Identification Tab
 - Detector results are not passed to Signal Conditioner automatically
 - Signal Conditioner flow graphs only work with IQ files, needs live radio capture support
 - Signal Conditioner only works with bursty signals and needs more isolation techniques beyond those related to amplitude thresholds
 - Feature Extractor needs to implement better types of statistical features to characterize signals
 - Signal Classifier needs additional experimentation with more signals and target devices to produce useful machine learning models
 - Signal Classifier needs to aggregate SOI data/results (files, statistics, classification results) and export it into Protocol Discovery/FISSURE library



Operational Flow

Protocol Discovery

- Protocol Discovery Tab – Identify RF protocol parameters for demodulation and perform analysis of the bitstream
 - Perform analysis on SOIs to measure RF parameters needed for demodulation and to help build the SOI profile
 - Distinguish between known and unknown signals based on confidence metrics
 - Attempt to demodulate signals and reduce bit errors by recursively loading flow graphs with updated input parameters
 - Look for patterns in the bitstream using pattern recognition and library entries for known protocols
 - Build custom dissectors for logging traffic
 - Apply analysis tools like CRC calculators and entropy plots as part of the automation but also have the option to use the tools manually
 - Add new protocol information to the FISSURE library



Operational Flow

Protocol Discovery – Current Limitations

- Protocol Discovery Tab
 - Needs a revamp to perform recursive demodulation and capture signal parameters/confidence metrics
 - Currently only loads demodulation flow graphs for known protocols
 - Need to define the steps in the process and what to do when information cannot be obtained
 - Need to integrate isolated manual analysis tools to be part of a larger semi-automated process
 - Need to pull SOIs from the FISSURE library and store protocol discovery results in the library
 - Need to create and apply bit analysis patterns for known protocols and message types



Operational Flow

Attacks

- Attack Tab – Execute single-stage, multi-stage, or fuzzing attacks using third-party tools, Python scripts, GNU Radio flow graphs, and crafted packets from the FISSURE library
 - Single-stage attacks: Python scripts and GNU Radio flow graphs that are loaded from a tree widget, change variable values before and during runtime, assign triggers, uses a simple header to list default variables
 - Multi-stage attacks: Execute multiple single-stage attacks in succession for specific durations, assign triggers, configure default variables, import/export
 - Fuzzing attacks: perform rudimentary data field fuzzing for messages in the FISSURE library (random/sequential limits, at regular intervals, recalculates CRC) or GNU Radio flow graph variable fuzzing for any flow graph
 - Packet crafter: view messages in the FISSURE library and construct binary/hex string data for importing/exporting, calculate CRCs, change values at the bit level, built-in Scapy injection tool for transmitting custom Wi-Fi messages



Operational Flow

Attacks – Current Limitations

- Attack Tab
 - No automation besides playlists
 - No advanced tipping and cueing system for staging attacks beyond simple triggers
 - No recommended actions based on protocol observations
 - No automation with sniffers to perform vulnerability analysis or react to state changes
 - Uses simple fuzzing mechanisms for known message types in the FISSURE library



Operational Flow

IQ Data

- IQ Data Tab – Perform manual analysis of signal data through recording, playback, live inspection, data manipulation, filtering, and viewing
 - Record IQ files using GNU Radio flow graphs and view the results in the data viewer, optional SigMF metadata support
 - Load IQ files for playback
 - View live inspection flow graphs for common operations or run flow graphs against a data file
 - Crop IQ files, convert data types, append IQ files, construct signals at regular timeslots, overlap signals, resample signals, normalize signals, strip leading/trailing data, split files
 - Generate OOK signals from input parameters
 - Plot IQ files, measure samples, zoom, plot ranges, apply filters (lowpass, bandpass), perform functions (magnitude, phase, instantaneous frequency, spectrogram, FFT, moving average, absolute value, differential, keep 1 in 2, unwrap, morse code deciphering, polar plot, load data in third-party programs like Gqrx or Inspectrum)



Operational Flow

IQ Data – Current Limitations

- IQ Data Tab
 - No live viewing of an IQ recording until it is completed
 - No annotations from metadata (e.g. SigMF) are displayed in the IQ data viewer
 - Limited IQ analysis and measurement capabilities
 - No signal to bit approximation for simple signals



Operational Flow

Archive

- Archive Tab – Download signal file collections from the AIS online archive, build signal playlists, construct datasets with randomizations for machine learning training
 - Download collections of IQ files recorded at different sample rates with SigMF metadata, individual files or entire collections
 - Build simple signal playlists of IQ files for replay transmitted from a single radio - specify sample rate, gain, channel, duration
 - Import IQ files and apply transformations to change the characteristics of signals for building machine learning models
 - Add noise, scale the amplitude, apply phase rotation, shift the frequency



Operational Flow

Archive – Current Limitations

- Archive Tab
 - Limited number of example signals in the online archive representing types of RF protocols, emitters, collection hardware, data formats, and environments
 - Dataset builder could use more advanced signal transformations to add more diversity
 - Dataset builder needs to generate metadata files to accurately describe the newly created data and the transformations applied



Operational Flow

Sensor Nodes

- Sensor Nodes Tab – Construct autorun playlists from single-stage and multi-stage attacks to run on remote sensor nodes, perform file transfer functions between the remote sensor nodes and the local workstation
 - Construct autorun playlists that can be exported to a file or transferred directly to a sensor node
 - Autorun playlists are comprised of single-stage and multi-stage attacks
 - Options to repeat, delay, apply timeouts to the playlist as whole and individual playlist items
 - Apply various types of acoustic, environmental, filesystem, networking, RF, time, and visual triggers for executing a playlist
 - Download files and folders from the remote sensor node
 - Transfer files to select folders on the remote sensor node



Operational Flow

Sensor Nodes – Current Limitations

- Sensor Nodes Tab
 - Triggers for individual archive playlist items
 - Coordination between nodes: networking, simultaneous actions
 - Full file navigation and secure data transfer
 - Autorun playlists cannot be configured to run a second playlist or react to events from a playlist



Operational Flow

Library

- Library Tab – Browse/search the FISSURE library, view saved user images, add/remove: protocols, modulation types, packet types, signals of interest, statistics, demodulation flow graphs, attacks
 - Browse FISSURE library contents for GNU Radio 3.8/3.10 files, supporting FISSURE files, archive files, attack categories, attacks, inspection flow graphs, protocol information, demodulation flow graphs, modulation types, packet types, SOI data, triggers
 - View image gallery for reference images and material relating to RF protocols
 - Search the library for SOIs by frequency, bandwidth, modulation, continuity, data field/bit values
 - Remove/delete items from the FISSURE library
 - Add new protocols, modulation types, packet types, signals of interest, statistics, demodulation flow graphs, and attacks through the user interface



Operational Flow

Library– Current Limitations

- Library Tab
 - No plugin support for importing and exporting capabilities
 - No database for managing large amounts of data
 - No multi-user access or permissions structure
 - Limited backup capabilities



Hardware & RF Protocols

Hardware

Supported Radios

- USRP: X3xx, B2xx, B20xmini, USRP2, N2xx, X410
- HackRF
- RTL2832U
- 802.11 Adapters
- LimeSDR
- bladeRF, bladeRF 2.0 micro
- Open Sniffer
- PlutoSDR
- SDRplay: RSPduo, RSPdx
- GPS Receivers



Hardware

Supported Computers and Operating Systems

- Desktops, laptops, mini PCs (Beelink), single-board computers (Raspberry Pi, Orange Pi)
- Operating Systems
 - BackBox Linux 8 (amd64)
 - DragonOS FocalX (x86_64)
 - Kali 23.1 (x64)
 - KDE neon 5.25 (x64) (6.0 not tested)
 - Parrot Security 6.1 (amd64)
 - Raspberry Pi OS (bookworm)
 - Ubuntu 18.04 (x64) – deprecated
 - Ubuntu 20.04 (x64)
 - Ubuntu 22.04 (x64)
 - Ubuntu 22.04 ARM (Orange Pi)



RF Protocols

Protocols in FISSURE Library

- 802.11x
- Bluetooth Classic
- Bluetooth Low Energy
- Clapper Plus
- DECT
- DSRC
- Ethernet
- FM Radio
- Garage Door
- General
- HD Radio
- Mode S
- Morse Code
- RDS
- SimpliCI
- TPMS
- X10
- Z-Wave



RF Protocols

Categories in Tools Menu

- 433 MHz
- 802.11
- Aircraft
- AIS
- AM Radio
- Audio
- Bluetooth
- Data
- Development
- Filters
- FM Radio
- GPS
- GSM
- Ham Radio
- Hardware
- HD Radio
- LTE
- M17
- Mapping
- POCSAG
- Point-to-Point
- Radiosonde
- Railway
- RFID
- Satellite
- SSH
- SDR
- Trunked Radio
- TPMS
- V2V
- Video
- VHF
- Weather
- Z-Wave
- Zigbee



RF Protocols

Categories in Standalone Flow Graphs Menu Library

- 802.11
- ACARS
- Clapper Plus
- FM
- Garage Door
- GSM
- HD Radio
- J2497
- Morse Code
- Painting
- POCSAG
- RDS
- Tesla
- TPMS
- X10
- V2V
- Z-Wave
- Zigbee
- Antenna Test Tx/Rx
- Noise Source
- Frequency Translating



Attack & Trigger Examples

Attack Categories

Categories in FISSURE Attack Tree

- Denial of Service
 - Jamming
- Spoofing
- Sniffing/Snooping
- Probe Attacks
- Installation of Malware
- Other
- Fuzzing



Attack Examples

Existing Attacks in FISSURE Library

- 802.11x - ARP Poisoning
- 802.11x - Beacon Flood
- 802.11x - Beacon Frame
- 802.11x - CTS Frame
- 802.11x - DIR-815 Exploit
- 802.11x - Deauthentication Frame
- 802.11x - ICMP
- 802.11x - LAND
- 802.11x - Probe Request Frame
- 802.11x - Relay
- 802.11x - SYN Flood
- 802.11x - Smurf
- 802.11x - TP-Link Archer A7 Exploit
- 802.11x - TP-Link Archer A7 Reboot
- 802.11x - UDP Hole Punch Server
- 802.11x - UDP Replay PCAP
- 802.11x - wifi_tx ARP Poisoning
- 802.11x - wifi_tx Beacon
- 802.11x - wifi_tx ICMP
- 802.11x - wifi_tx LAND
- 802.11x - wifi_tx SYN Flood
- 802.11x - wifi_tx Smurf
- 802.11x - wifi_tx UDP ARP Poisoning
- 802.11x - wifi_tx UDP Crafter
- 802.11x - wifi_tx UDP Crafter QoS
- 802.11x - wifi_tx UDP File
- 802.11x - wifi_tx UDP Mapper
- 802.11x - wifi_tx UDP PCAP
- Bluetooth Classic - btrx
- Bluetooth Low Energy - ble_dump
- Bluetooth Low Energy - btrx
- Clapper Plus - Transmit
- DECT - Audio Sink
- DECT - CW
- DECT - File Sink
- DECT - File Sink IQ
- DECT - From File
- DECT - Gaussian
- DECT - Simple Replay
- DECT - Wireshark
- DSRC - From File
- DSRC - wifi_tx Generator
- Ethernet - DIR-815 Exploit
- Ethernet - MAC Flooding
- Ethernet - TP-Link Archer A7 Exploit
- FM Radio - Audio Sink
- FM Radio - CW
- FM Radio - File Sink
- FM Radio - From File
- FM Radio - From Wav File
- FM Radio - Gaussian
- FM Radio - Simple Replay
- Garage Door - Cycle
- Garage Door - Transmit
- General - IQ Playback Repeat
- General - IQ Playback Single
- General - IQ Record
- General - IQ Record Squelch
- General - Instantaneous Frequency
- General - OOK Transmit
- General - Sleep
- HD Radio - Naughty String Injection
- Mode S - Dump1090 Interactive
- Mode S - Dump1090 Raw File
- Mode S - Dump1090 Raw Network
- Mode S - Fields
- Mode S - From File
- Mode S - dump978 uat2text
- Mode S - gr-adsb Brief
- Mode S - gr-adsb To Webserver
- Mode S - gr-adsb Verbose
- Mode S - gr-adsb Webserver
- Morse Code - File
- Morse Code - Transmit
- RDS - Fields
- RDS - File Source
- SimpliCI - Access Point DoS
- SimpliCI - Add Node
- SimpliCI - CW
- SimpliCI - Fields
- SimpliCI - From File
- SimpliCI - Gaussian
- SimpliCI - Unlink Node
- TPMS - Receive
- TPMS - Transmit
- X10 - Decode
- X10 - Fields
- X10 - Off
- X10 - On
- ZWAVE - Color List
- ZWAVE - Off
- ZWAVE - On
- ZWAVE - Random Colors
- ZWAVE - Receive
- ZWAVE - Transmit



Trigger Examples

Simultaneous Triggers to Kick off Attacks & Playlists

Category	Trigger Name	Hardware/Requirements	Description
Acoustic	Sound Threshold	Microphone	Activates once a custom audio threshold is reached on the active computer microphone
Environmental	Sunrise/Sunset	Internet Connection	Checks for sunrise and sunset time using “%S” and “%s” returns from wttr.in
Environmental	Temperature	Internet Connection	Compares the temperature in a location to a value using wttr.in
Environmental	Weather	Internet Connection	Checks for rain, snow/sleet, clear, cloudy/fog in a city using keywords found in the “%C” returns from wttr.in
Environmental	Wind	Internet Connection	Checks for wind speeds above a threshold using “%w” returns from wttr.in
Filesystem	File Modified	None	Waits for a file to be modified
Filesystem	Folder Modified	None	Waits for a folder to change the number of files it contains



Trigger Examples

Simultaneous Triggers to Kick off Attacks & Playlists

Category	Trigger Name	Hardware/Requirements	Description
Networking	Webserver Curl	Network Connection	Creates a local webserver that listens for a <i>curl</i> <code>http://<ip_address>:<port></code> command to exit
RF	Cellular Tower	RTL2832U	Continuously runs “CellSearch” at a specific frequency to find a matching PCI/cell ID for a site
RF	Detect SSID	WLAN Interface (Managed Mode)	Looks for a matching SSID name using <i>iwlist</i> for a provided interface
RF	GPS Line	GPS Receiver	Checks if current GPS coordinates have crossed either a latitude or longitude threshold (N-S/E-W line)
RF	GPS Point	GPS Receiver	Checks current GPS coordinates until they are a specified distance from a point
RF	Plane Spotting	RTL2832U	Looks for an ICAO in dump1090 stdout from an RTL dongle



Trigger Examples

Simultaneous Triggers to Kick off Attacks & Playlists

Category	Trigger Name	Hardware/Requirements	Description
RF	Power Threshold	USRP B2x0	Looks for a signal above a power threshold at a specified frequency. Reuses the TSI fixed detector flow graphs without the GUI.
RF	RDS Keyword	USRP B2x0	Looks for a keyword in the RDS printed from gr-rds for a radio station
RF	X10 Demod	USRP B2x0	Demodulates X10 signals for a USRP B210 and checks the flow graph stdout for matching text
Time	Sensor Node Time	None	Waits for date/time to elapse
Time	Timer	None	Waits for N seconds after trigger script start
Visual	Motion Detector	Camera	Checks for motion using the default camera





Learn More

- FISSURE

- <https://github.com/ainfosec/FISSURE>
- <https://www.ainfosec.com/fissure>
- <https://www.youtube.com/@assuredinformationsecurity>

- AIS

- <https://www.ainfosec.com>

Chris Poore

Senior Reverse Engineer

poorec@ainfosec.com

Assured Information Security, Inc.

Christine Todd

Program Manager – SAE

toddc@ainfosec.com

Assured Information Security, Inc.

